

Topic 1.6 - Polynomial Functions and End Behavior

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

End behavior describes the global direction of a polynomial far to the left and right. It is controlled by the first nonzero term of greatest degree, not by intercepts or middle coefficients. Careful parameter analysis is essential when a nominal leading term can cancel.

Learning targets

- Determine end behavior from degree parity and leading coefficient.
- Express end behavior with words, arrows, and limit notation and justify leading-term dominance.
- Analyze parameter families in which the degree and end behavior change through cancellation.

Language and structure

Term	Meaning in this topic
leading term	The nonzero term with greatest exponent.
leading coefficient	The coefficient of the leading term.
even-degree behavior	Both ends move in the same vertical direction.
odd-degree behavior	The two ends move in opposite directions.
leading-term dominance	Lower-degree terms become small relative to the leading term as $ x $ grows.

Launch - reason before calculating

Without graphing, determine the end behavior of $P(x) = -3x^5 + 2x^2 - 1$. Explain why the $2x^2$ term cannot make both ends rise.

Concept 1: Parity and leading coefficient

Core idea

Positive even leading terms rise on both ends; negative even leading terms fall on both ends. Positive odd leading terms fall left and rise right; negative odd leading terms rise left and fall right. Zeros and vertical shifts do not alter this global classification.

Worked example - annotate the reasoning

Problem. Classify $F(x) = 0.2x^6 - 4x^3 + 7$.

Model reasoning. The leading term $0.2x^6$ is positive even, so $F(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

Check your understanding

Independent

No calculator

Classify $G(x) = -5x^8 + 100x^7 - 1$.

Concept 2: Limit notation and dominance

Core idea

Factor the leading term to justify dominance: $P(x) = a_n x^n [1 + \frac{a_{n-1}}{a_n x} + \dots]$. The bracket tends to 1, so the sign and parity of $a_n x^n$ govern the limits. This is a proof, not merely a graph pattern.

Worked example - annotate the reasoning

Problem. Justify the right-end behavior of $H(x) = 4x^4 - 300x^3 + 2$.

Model reasoning. $H = 4x^4[1 - 75/x + 1/(2x^4)]$. The bracket tends to 1 as $x \rightarrow \infty$, so $H \rightarrow \infty$.

Check your understanding

Independent

No calculator

Write both end-behavior limits for $K(x) = 7 - 2x^3 + x$.

Concept 3: Parameter cancellation changes degree

Core idea

In a parameter family, determine the actual first nonzero coefficient before classifying end behavior. A special parameter value may cancel the expected leading term, exposing a lower-degree term with different parity or sign.

Worked example - annotate the reasoning

Problem. Classify $F_k(x) = (k - 2)x^5 + (3k + 1)x^4 - 7x + 9$ for $k = 2$ and $k \neq 2$.

Model reasoning. If $k = 2$, the x^5 term cancels and the leading term is $7x^4$, so both ends rise. If $k \neq 2$, degree 5 is odd; the right end follows the sign of $k - 2$ and the left end is opposite.

Check your understanding

Independent

No calculator

For $Q_a(x) = (a + 1)x^6 + (2 - a)x^5 + x^2$, find all a giving left end down and right end up.

Representation laboratory

Connect at least two representations

A graph rises on both far ends and has three visible real zeros. List all possible parities for its degree, state the sign of its leading coefficient, and explain why three intercepts do not determine the exact degree.

Challenge laboratory

Challenge - justify every claim

Find all parameter values a, b for which $P(x) = (a - 1)x^6 + (b + 2)x^5 - 4x^2 + 1$ has left end down and right end up. State whether the answer is unique.

Error clinic and calculator strategy

Common errors to diagnose

- Using the constant term or y -intercept to classify end behavior.
- Assuming the largest numerical coefficient creates the leading term.
- Ignoring cancellation at a special parameter value.
- Confusing end behavior with increasing/decreasing behavior on the whole real line.

Technology decision

A graphing window may not extend far enough for the leading term to dominate, especially when lower-degree coefficients are large. Use algebra first, then expand the window to confirm. Contextual models should be graphed only on their intended domain as well as globally for structural comparison.

AP-style synthesis**Multi-part reasoning task**

Let $P_a(x) = (a - 3)x^6 + (2a + 1)x^5 - 7x + 4$.

A. Classify end behavior for $a > 3$ and $a < 3$.

B. Determine the end behavior when $a = 3$.

C. Find every a for which the ends move in opposite directions and justify uniqueness.

Exit ticket**Before you leave**

1. State the four parity/sign end-behavior cases.

2. Why can a calculator window misrepresent end behavior?

3. What must be checked first in a parameter family?
